

# Random Forests and Gradient Boosting

“Bagging” and “Boosting”

# The Bootstrap Sample and Bagging

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Simple ideas to improve any model via ensemble

# Bootstrap Samples

- Random samples of your data *with replacement* that are the same size as original data.
- Some observations will not be sampled. These are called *out-of-bag observations*

**Example:** Suppose you have 10 observations, labeled 1-10

| Bootstrap Sample Number | Training Observations  | Out-of-Bag Observations |
|-------------------------|------------------------|-------------------------|
| 1                       | {1,3,2,8,3,6,4,2,8,7}  | {5,9,10}                |
| 2                       | {9,1,10,9,7,6,5,9,2,6} | {3,4,8}                 |
| 3                       | {8,10,5,3,8,9,2,3,7,6} | {1,4}                   |

# Bootstrap Samples

- Can be proven that a bootstrap sample will contain approximately 63% of the observations.
- The sample size is the same as the original data as some observations are repeated.
- Some observations left out of the sample (~37% out-of-bag)
- Uses:
  - Alternative to traditional validation/cross-validation
  - **Create Ensemble Models using different training sets (Bagging)**

# Bagging

(Bootstrap Aggregating)

- Let  $k$  be the number of bootstrap samples
- For each bootstrap sample, create a classifier using that sample as training data
  - Results in  $k$  different models
- Ensemble those classifiers
  - A test instance is assigned to the class that received the highest number of votes.

# Bagging Example

input variable

target

|   |     |     |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| x | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 | 1.0 |
| y | 1   | 1   | 1   | -1  | -1  | -1  | -1  | 1   | 1   | 1   |

- 10 observations in original dataset
- Suppose we build a decision tree with only 1 split.
- The best accuracy we can get is 70%
  - Split at  $x=0.35$
  - Split at  $x=0.75$
- A tree with one split called a **decision stump**

# Bagging Example

Let's see how bagging might improve this model:

1. Take 10 Bootstrap samples from this dataset.
2. Build a decision stump for each sample.
3. Aggregate these rules into a voting ensemble.
4. Test the performance of the voting ensemble on the whole dataset.

# Bagging Example

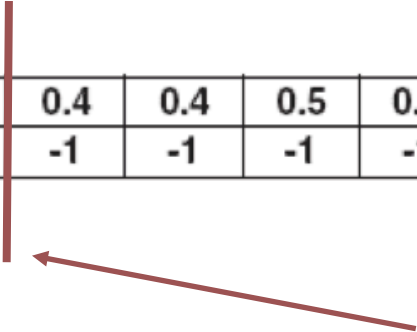
## Classifier 1

Bagging Round 1:

|   |     |     |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| x | 0.1 | 0.2 | 0.2 | 0.3 | 0.4 | 0.4 | 0.5 | 0.6 | 0.9 | 0.9 |
| y | 1   | 1   | 1   | 1   | -1  | -1  | -1  | -1  | 1   | 1   |

$x \leq 0.35 \implies y = 1$

$x > 0.35 \implies y = -1$



Best decision stump splits  
at  $x=0.35$

First bootstrap sample:

Some observations chosen multiple times.

Some not chosen.



# Bagging Example

## Classifiers 1-5

Bagging Round 1:

|   |     |     |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| x | 0.1 | 0.2 | 0.2 | 0.3 | 0.4 | 0.4 | 0.5 | 0.6 | 0.9 | 0.9 |
| y | 1   | 1   | 1   | 1   | -1  | -1  | -1  | -1  | 1   | 1   |

$x \leq 0.35 \implies y = 1$

$x > 0.35 \implies y = -1$

Bagging Round 2:

|   |     |     |     |     |     |     |     |   |   |   |
|---|-----|-----|-----|-----|-----|-----|-----|---|---|---|
| x | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.8 | 0.9 | 1 | 1 | 1 |
| y | 1   | 1   | 1   | -1  | -1  | 1   | 1   | 1 | 1 | 1 |

$x \leq 0.65 \implies y = 1$

$x > 0.65 \implies y = 1$

Bagging Round 3:

|   |     |     |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| x | 0.1 | 0.2 | 0.3 | 0.4 | 0.4 | 0.5 | 0.7 | 0.7 | 0.8 | 0.9 |
| y | 1   | 1   | 1   | -1  | -1  | -1  | -1  | -1  | 1   | 1   |

$x \leq 0.35 \implies y = 1$

$x > 0.35 \implies y = -1$

Bagging Round 4:

|   |     |     |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| x | 0.1 | 0.1 | 0.2 | 0.4 | 0.4 | 0.5 | 0.5 | 0.7 | 0.8 | 0.9 |
| y | 1   | 1   | 1   | -1  | -1  | -1  | -1  | -1  | 1   | 1   |

$x \leq 0.3 \implies y = 1$

$x > 0.3 \implies y = -1$

Bagging Round 5:

|   |     |     |     |     |     |     |     |   |   |   |
|---|-----|-----|-----|-----|-----|-----|-----|---|---|---|
| x | 0.1 | 0.1 | 0.2 | 0.5 | 0.6 | 0.6 | 0.6 | 1 | 1 | 1 |
| y | 1   | 1   | 1   | -1  | -1  | -1  | -1  | 1 | 1 | 1 |

$x \leq 0.35 \implies y = 1$

$x > 0.35 \implies y = -1$

# Bagging Example

## Classifiers 6-10

Bagging Round 6:

|   |     |     |     |     |     |     |     |     |     |   |                               |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|---|-------------------------------|
| x | 0.2 | 0.4 | 0.5 | 0.6 | 0.7 | 0.7 | 0.7 | 0.8 | 0.9 | 1 | $x \leq 0.75 \implies y = -1$ |
| y | 1   | -1  | -1  | -1  | -1  | -1  | -1  | 1   | 1   | 1 | $x > 0.75 \implies y = 1$     |

Bagging Round 7:

|   |     |     |     |     |     |     |     |     |     |   |                               |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|---|-------------------------------|
| x | 0.1 | 0.4 | 0.4 | 0.6 | 0.7 | 0.8 | 0.9 | 0.9 | 0.9 | 1 | $x \leq 0.75 \implies y = -1$ |
| y | 1   | -1  | -1  | -1  | -1  | 1   | 1   | 1   | 1   | 1 | $x > 0.75 \implies y = 1$     |

Bagging Round 8:

|   |     |     |     |     |     |     |     |     |     |   |                               |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|---|-------------------------------|
| x | 0.1 | 0.2 | 0.5 | 0.5 | 0.5 | 0.7 | 0.7 | 0.8 | 0.9 | 1 | $x \leq 0.75 \implies y = -1$ |
| y | 1   | 1   | -1  | -1  | -1  | -1  | -1  | 1   | 1   | 1 | $x > 0.75 \implies y = 1$     |

Bagging Round 9:

|   |     |     |     |     |     |     |     |     |   |   |                               |
|---|-----|-----|-----|-----|-----|-----|-----|-----|---|---|-------------------------------|
| x | 0.1 | 0.3 | 0.4 | 0.4 | 0.6 | 0.7 | 0.7 | 0.8 | 1 | 1 | $x \leq 0.75 \implies y = -1$ |
| y | 1   | 1   | -1  | -1  | -1  | -1  | -1  | 1   | 1 | 1 | $x > 0.75 \implies y = 1$     |

Bagging Round 10:

|   |     |     |     |     |     |     |     |     |     |     |                               |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------------------------------|
| x | 0.1 | 0.1 | 0.1 | 0.1 | 0.3 | 0.3 | 0.8 | 0.8 | 0.9 | 0.9 | $x \leq 0.05 \implies y = -1$ |
| y | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | $x > 0.05 \implies y = 1$     |

# Bagging Example

## Predictions from each Classifier

| Round      | x=0.1 | x=0.2 | x=0.3 | x=0.4 | x=0.5 | x=0.6 | x=0.7 | x=0.8 | x=0.9 | x=1.0 |
|------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1          | 1     | 1     | 1     | -1    | -1    | -1    | -1    | -1    | -1    | -1    |
| 2          | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     |
| 3          | 1     | 1     | 1     | -1    | -1    | -1    | -1    | -1    | -1    | -1    |
| 4          | 1     | 1     | 1     | -1    | -1    | -1    | -1    | -1    | -1    | -1    |
| 5          | 1     | 1     | 1     | -1    | -1    | -1    | -1    | -1    | -1    | -1    |
| 6          | -1    | -1    | -1    | -1    | -1    | -1    | -1    | 1     | 1     | 1     |
| 7          | -1    | -1    | -1    | -1    | -1    | -1    | -1    | 1     | 1     | 1     |
| 8          | -1    | -1    | -1    | -1    | -1    | -1    | -1    | 1     | 1     | 1     |
| 9          | -1    | -1    | -1    | -1    | -1    | -1    | -1    | 1     | 1     | 1     |
| 10         | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     |
| Sum        | 2     | 2     | 2     | -6    | -6    | -6    | -6    | 2     | 2     | 2     |
| Sign       | 1     | 1     | 1     | -1    | -1    | -1    | -1    | 1     | 1     | 1     |
| True Class | 1     | 1     | 1     | -1    | -1    | -1    | -1    | 1     | 1     | 1     |

Ensemble Classifier has 100% Accuracy

# Bagging Summary

- Improves generalization error on models with high variance
- Bagging helps reduce errors associated with random fluctuations in training data (high variance)
- If base classifier is stable (not suffering from high variance), bagging can actually make it worse
- Bagging does not focus on any particular observations in the training data (unlike boosting)

# Random Forests

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Tin Kam Ho (1995, 1998)

Leo Breiman (2001)

# Random Forests

- Random Forests are ensembles of decision trees similar to the one we just saw
- Ensembles of decision trees work best when their predictions are not correlated – they each find different patterns in the data
- Problem: Bagging tends to create correlated trees
- Two Solutions: (a) Randomly subset features considered for each split. (b) Use unpruned decision trees in the ensemble.

# Random Forests

- A collection of unpruned decision or regression trees.
- Each tree is build on a bootstrap sample of the data **and** a subset of features are considered at each split.
  - The number of features considered for each split is a parameter called *mtry*.
  - Brieman (2001) suggests  $mtry = \sqrt{p}$  where  $p$  is the number of features
  - I'd suggest setting *mtry* equal to 5-10 values evenly spaced between 2 and  $p$  and choosing the parameter by validation
  - Overall, the model is relatively insensitive to values for *mtry*.
- The results from the trees are ensembled into one voting classifier.

# Random Forests

## Summary

### ➤ Advantages

- Computationally Fast – can handle thousands of input variables
- Trees can be trained simultaneously
- Exceptional Classifiers – one of most accurate available
- Provide information on variable importance for the purposes of feature selection
- Can effectively handle missing data

### ➤ Disadvantages

- No interpretability in final model aside from variable importance
- Prone to overfitting
- Lots of tuning parameters like the number of trees, the depth of each tree, the percentage of variables passed to each tree



# Boosting

...

# Boosting Overview

- Like bagging, going to draw a sample of the observations from our data with replacement
- Unlike bagging, the observations not sampled randomly
- Boosting assigns a weight to each training observation and uses that weight as a sampling distribution
  - Higher weight observations more likely to be chosen.
- May adaptively change that weight in each round
- **The weight is higher for examples that are harder to classify**

# Boosting Example

input variable

|   |     |     |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| x | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 | 1.0 |
| y | 1   | 1   | 1   | -1  | -1  | -1  | -1  | 1   | 1   | 1   |

target

- Same dataset used to illustrate bagging
- Boosting typically requires fewer rounds of sampling and classifier training.
- Start with equal weights for each observation
- Update weights each round based on the classification errors

# Boosting Example

Boosting Round 1:

|   |     |     |     |     |     |     |     |     |     |   |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|---|
| x | 0.1 | 0.4 | 0.5 | 0.6 | 0.6 | 0.7 | 0.7 | 0.7 | 0.8 | 1 |
| y | 1   | -1  | -1  | -1  | -1  | -1  | -1  | -1  | 1   | 1 |

Boosting Round 2:

|   |     |     |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| x | 0.1 | 0.1 | 0.2 | 0.2 | 0.2 | 0.2 | 0.3 | 0.3 | 0.3 | 0.3 |
| y | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   |

Boosting Round 3:

|   |     |     |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| x | 0.2 | 0.2 | 0.4 | 0.4 | 0.4 | 0.4 | 0.5 | 0.6 | 0.6 | 0.7 |
| y | 1   | 1   | -1  | -1  | -1  | -1  | -1  | -1  | -1  | -1  |

(a) Training records chosen during boosting

| Round | x=0.1 | x=0.2 | x=0.3 | x=0.4 | x=0.5 | x=0.6 | x=0.7 | x=0.8 | x=0.9 | x=1.0 |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1     | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   |
| 2     | 0.311 | 0.311 | 0.311 | 0.01  | 0.01  | 0.01  | 0.01  | 0.01  | 0.01  | 0.01  |
| 3     | 0.029 | 0.029 | 0.029 | 0.228 | 0.228 | 0.228 | 0.228 | 0.009 | 0.009 | 0.009 |

(b) Weights of training records

# Boosting: Weighted Ensemble

- Unlike Bagging, Boosted Ensembles usually weight the votes of each classifier by a function of their accuracy.
- If a classifier gets the higher weight observations wrong, it has a higher error rate.
- More accurate classifiers get higher weight in the prediction.

# Boosting: Classifier weights

Errors made: First 3 observations

|   |     |     |     |     |     |     |     |     |     |   |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|---|
| x | 0.1 | 0.4 | 0.5 | 0.6 | 0.6 | 0.7 | 0.7 | 0.7 | 0.8 | 1 |
| y | 1   | -1  | -1  | -1  | -1  | -1  | -1  | -1  | 1   | 1 |

Errors made: Middle 4 observations

|   |     |     |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| x | 0.1 | 0.1 | 0.2 | 0.2 | 0.2 | 0.2 | 0.3 | 0.3 | 0.3 | 0.3 |
| y | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   |

Errors made: Last 3 observations

|   |     |     |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| x | 0.2 | 0.2 | 0.4 | 0.4 | 0.4 | 0.4 | 0.5 | 0.6 | 0.6 | 0.7 |
| y | 1   | 1   | -1  | -1  | -1  | -1  | -1  | -1  | -1  | -1  |

(a) Training records chosen during boosting

| Round | x=0.1 | x=0.2 | x=0.3 | x=0.4 | x=0.5 | x=0.6 | x=0.7 | x=0.8 | x=0.9 | x=1.0 |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1     | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   |
| 2     | 0.311 | 0.311 | 0.311 | 0.01  | 0.01  | 0.01  | 0.01  | 0.01  | 0.01  | 0.01  |
| 3     | 0.029 | 0.029 | 0.029 | 0.228 | 0.228 | 0.228 | 0.228 | 0.009 | 0.009 | 0.009 |

(b) Weights of training records

# Boosting: Classifier weights

Errors made: First 3 observations

|   |     |     |     |     |     |     |     |     |     |   |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|---|
| x | 0.1 | 0.4 | 0.5 | 0.6 | 0.6 | 0.7 | 0.7 | 0.7 | 0.8 | 1 |
| y | 1   | -1  | -1  | -1  | -1  | -1  | -1  | -1  | 1   | 1 |

Errors made: Middle 4 observations

|   |     |     |     |     |     |     |     |     |     |     |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| x | 0.1 | 0.1 | 0.2 | 0.2 | 0.2 | 0.2 | 0.3 | 0.3 | 0.3 | 0.3 |
| y | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   |

Errors made: Last 3 observations

|   |     |     |     |     |     |     |  |  |  |  |
|---|-----|-----|-----|-----|-----|-----|--|--|--|--|
| x | 0.2 | 0.2 | 0.4 | 0.4 | 0.4 | 0.4 |  |  |  |  |
| y | 1   | 1   | -1  | -1  | -1  | -1  |  |  |  |  |

Lowest weighted error.  
Highest weighted model.

(a) Training records chosen during boosting

| Round | x=0.1 | x=0.2 | x=0.3 | x=0.4 | x=0.5 | x=0.6 | x=0.7 | x=0.8 | x=0.9 | x=1.0 |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1     | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   |
| 2     | 0.311 | 0.311 | 0.311 | 0.01  | 0.01  | 0.01  | 0.01  | 0.01  | 0.01  | 0.01  |
| 3     | 0.029 | 0.029 | 0.029 | 0.228 | 0.228 | 0.228 | 0.228 | 0.009 | 0.009 | 0.009 |

(b) Weights of training records

# Boosting: Weighted Ensemble

| Round | Split Point | Left Class | Right Class | Weight |
|-------|-------------|------------|-------------|--------|
| 1     | 0.75        | -1         | 1           | 1.738  |
| 2     | 0.05        | 1          | 1           | 2.7784 |
| 3     | 0.3         | 1          | -1          | 4.1195 |

Classifier Decision Rules and Classifier Weights

| Round | x=0.1 | x=0.2 | x=0.3 | x=0.4 | x=0.5 | x=0.6 | x=0.7 | x=0.8 | x=0.9 | x=1.0 |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1     | -1    | -1    | -1    | -1    | -1    | -1    | -1    | 1     | 1     | 1     |
| 2     | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     |
| 3     | 1     | 1     | 1     | -1    | -1    | -1    | -1    | -1    | -1    | -1    |
| Sum   | 5.16  | 5.16  | 5.16  | -3.08 | -3.08 | -3.08 | -3.08 | 0.397 | 0.397 | 0.397 |
| Sign  | 1     | 1     | 1     | -1    | -1    | -1    | -1    | 1     | 1     | 1     |

Individual Classifier Predictions and Weighted Ensemble Predictions



# Boosting: Weighted Ensemble

| Round | Split Point | Left Class | Right Class | Weight |
|-------|-------------|------------|-------------|--------|
| 1     | 0.75        | -1         | 1           | 1.738  |
| 2     | 0.05        | 1          | 1           | 2.7784 |
| 3     | 0.3         | 1          | -1          | 4.1195 |

Classifier Decision Rules and Classifier Weights

| Round | x=0.1 | x=0.2 | x=0.3 | x=0.4 | x=0.5 | x=0.6 | x=0.7 | x=0.8 | x=0.9 | x=1.0 |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1     | -1    | -1    | -1    | -1    | -1    | -1    | -1    | 1     | 1     | 1     |
| 2     | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     |
| 3     | 1     | 1     | 1     | 1     | 1     | 1     | -1    | -1    | -1    | -1    |
| Sum   | 5.16  | 5.16  | 5.16  | -3.08 | -3.08 | -3.08 | -3.08 | 0.397 | 0.397 | 0.397 |
| Sign  | 1     | 1     | 1     | -1    | -1    | -1    | -1    | 1     | 1     | 1     |

$$5.16 = -1.738 + 2.7784 + 4.1195$$

Individual Classifier Predictions and Weighted Ensemble Predictions

# (Major) Boosting Algorithms

AdaBoost (This is sooo 2007)

Gradient Boosting [xgboost]  
(Welcome to the New Age of learning)

# (Self-Study)

## AdaBoost Details: The Classifier Weights

- Let  $w_j$  be the weight of observation  $j$  entering into present round.
- Let  $m_j = 1$  if observation  $j$  is misclassified, 0 otherwise
- The error of the classifier this round is

$$\epsilon_i = \frac{1}{N} \sum_{j=1}^N w_j m_j$$

- The voting weight for the classifier this round is then

$$\alpha_i = \frac{1}{2} \ln \left( \frac{1 - \epsilon_i}{\epsilon_i} \right)$$

# (Self-Study)

## AdaBoost Details: Updating observation Weights

To update the observation weights from the current round (round  $i$ ) to the next round (round  $i + 1$ ):

$$w_j^{(i+1)} = w_j^i e^{-\alpha_j} \quad \text{if observation } j \text{ was correctly classified}$$
$$w_j^{(i+1)} = w_j^i e^{\alpha_j} \quad \text{if observation } j \text{ was misclassified}$$

The new weights are then normalized to sum to 1 so they form a probability distribution.

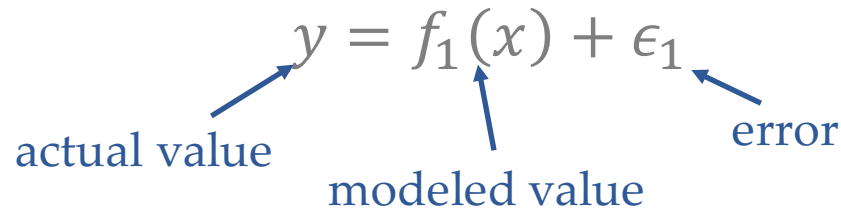
# Gradient Boosting

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The latest and greatest  
(Jerome H. Friedman 1999)

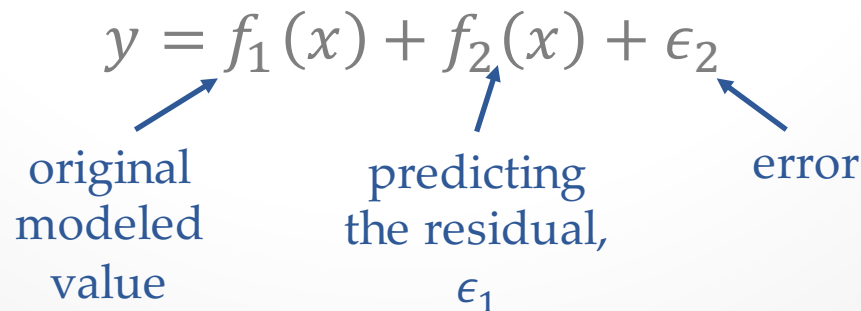
# Gradient Boosting Overview

- Build a simple model  $f_1(x)$  trying to predict a target  $y$
- It has error, right?

$$y = f_1(x) + \epsilon_1$$


A diagram showing the equation  $y = f_1(x) + \epsilon_1$ . Three blue arrows point from text labels to parts of the equation: 'actual value' points to  $y$ , 'modeled value' points to  $f_1(x)$ , and 'error' points to  $\epsilon_1$ .

- Now, let's try to predict that error with another simple model,  $f_2(x)$ . Unfortunately, it still has some error:

$$y = f_1(x) + f_2(x) + \epsilon_2$$


A diagram showing the equation  $y = f_1(x) + f_2(x) + \epsilon_2$ . Three blue arrows point from text labels to parts of the equation: 'original modeled value' points to  $f_1(x)$ , 'predicting the residual,  $\epsilon_1$ ' points to  $f_2(x)$ , and 'error' points to  $\epsilon_2$ .

# Gradient Boosting Overview

- We could just continue to add model after model, trying to predict the residuals from the previous set of models.

$$y = f_1(x) + f_2(x) + f_3(x) + \cdots + f_k(x) + \epsilon_k$$

The diagram illustrates the components of the gradient boosting equation. Arrows point from descriptive text below to specific terms in the equation above:

- An arrow points from "original modeled value" to  $f_1(x)$ .
- An arrow points from "predicting the residual,  $\epsilon_1$ " to  $f_2(x)$ .
- An arrow points from "predicting the residual,  $\epsilon_2$ " to  $f_3(x)$ .
- An arrow points from "presumably very small error" to  $\epsilon_k$ .

# Gradient Boosting Overview

- To address the obvious problem of overfitting, we'll dampen the effect of the additional models by only taking a “step” toward the solution in that direction.
- We'll also start (in continuous problems) with a constant function (intercept)
- The **step-sizes** are automatically determined at each round inside the method

$$y = \gamma_1 + \gamma_2 f_2(x) + \gamma_3 f_3(x) + \cdots + \gamma_k f_k(x) + \epsilon_k$$



# Gradient Boosted Trees

- Gradient boosting yields a additive ensemble model
- The key to gradient boosting is using “**weak learners**”
  - Typically simple, shallow decision/regression trees
  - Computationally fast and efficient
  - Alone, make poor predictions but ensembled in this additive fashion provide superior results

# Gradient Boosting and Overfitting

- In general, the "step-size" is not enough to prevent us from overfitting the training data
- To further aid in this mission, we must use some form of **regularization** to prevent overfitting:
  1. Control the number of trees/classifiers used in the prediction
    - Larger number of trees => More prone to overfitting
    - Choose a number of trees by observing out-of-sample error
  2. Use a *shrinkage parameter* ("learning rate") to effectively lessen the step-size taken at each step. Often called eta,  $\eta$ 
    - $y = \gamma_1 + \eta \gamma_2 f_2(x) + \eta \gamma_3 f_3(x) + \dots + \eta \gamma_k f_k(x) + \epsilon_k$
    - Smaller values of eta => Less prone to overfitting
    - eta = 1 => no regularization

# Gradient Boosting Summary

## ➤ Advantages

- Exceptional model – one of most accurate available, generally superior to Random Forests when well trained
- Can provide information on variable importance for the purposes of variable selection

## ➤ Disadvantages

- Model lacks interpretability in the classical sense aside from variable importance
- The trees must be trained sequentially so computationally this method is slower than Random Forest
- Extra tuning parameter over Random Forests, the regularization or shrinkage parameter,  $\eta$ .

# Notes about EM

- EM has node for Random Forest (HP tab=> HP Forest)
  - Uses CHAID unlike other implementations
  - Does not perform bootstrap sampling
  - Does not appear to work as well as the randomForest package in R
- EM has node for gradient boosting
  - Personally I recommend the "extreme gradient boosting" implementation of this method, which is called *xgboost* both in R and python.
  - This implementation appears to be stronger and faster than the one in SAS